Muhammet Celal Gokalp

Product Feature Owner | ADAS Systems Engineer

+44 07397791625 | muhammetcelalgokalp@gmail.com | Coventry, CV5 8HN, UK | British, Turkish
linkedin.com/in/celal-gökalp-4616ab122

PROFESSIONAL SUMMARY

Product Feature Owner and ADAS Systems Engineer with extensive experience taking end-to-end ownership of active safety features from concept to production. Proven track record embedded at a major automotive OEM (Jaguar Land Rover) as the central liaison between OEM stakeholders, suppliers, and engineering teams, translating customer requirements into feature specifications and key performance requirements. Hands-on in ASPICE-aligned V-model development, proving ground testing, and safety-driven requirements (ISO 26262, SOTIF, UN Reg 79). Currently validating an L4 autonomous driving stack at FiveAI / Bosch, developing ADAS and Euro NCAP scenarios and building AI-assisted tooling to scale test automation.

WORK EXPERIENCE

Lead Application Engineer (Solutions & Integration)

10/2021 – Present

FiveAI / Bosch

Cambridge, United Kingdom

- **ADAS Scenario Development & Evaluation:** Developed and evaluated test scenarios for ADAS features, including Automatic Lane Change on the ADA project, where the rules and scenarios were built directly for compliance with UN Regulation No. 79. Also created Euro NCAP junction-handling scenarios, using OpenSCENARIO 2 (OSC2) and evaluating results against strict validation and KPI criteria.
- **Simulation Ecosystem Leadership:** Founding expert of the Simulation EcoSystem team, driving cross-functional alignment on complex mapping integrations (SDB/XODR) and toolchain triage to ensure uninterrupted service continuity for enterprise customers.
- **Efficient AI Workflows & Tooling:** Leveraged AI workflows to streamline QA and test tooling, rapidly prototyping custom internal utilities (e.g., local STAX uploader, JSON parameter GUI) to improve test efficiency and support the core engineering team.
- **Data-Driven Product Alignment:** Evaluated thousands of high-level simulation results against strict validation and KPI criteria, engineering pipelines to route complex data to frontend analysis pages and translating system behaviours into actionable design improvements.
- **Automotive Standards Standardization:** Engineered modular EuroNCAP Simax test scenario templates (including crossing-path junction scenarios) for Core projects, bridging prior real-world ADAS testing expertise with virtual simulation pipelines to scale development capacity.

ADAS Feature Owner / Systems Engineer

12/2017 – 10/2021

AVL Powertrain UK

Coventry, United Kingdom

- **Feature Ownership & Delivery:** Embedded directly at Jaguar Land Rover as ADAS Feature Owner, taking end-to-end responsibility for Rear Cross Traffic Alert, Blind Spot Monitor, and Rear Pre-Crash, and acting as the primary technical bridge between AVL and OEM stakeholders.
- **Feature Design & Specification:** Designed ADAS features including Intelligent Adaptive Cruise Control, Lane Keep Assist, Highway Assist, and Lane Change Assist, translating customer specifications into use cases, key performance requirements, and system state diagrams in DOORS and MKS Integrity.
- **ASPICE & Functional Safety:** Worked within ASPICE-compliant V-model processes, decomposing high-level into low-level requirements with traceable test cases for each system state. Authored ISO 26262 functional safety requirements for a Lane Change feature.
- **Homologation:** Contributed directly to the vehicle homologation process for the Rear Cross Traffic Alert feature, ensuring system performance metrics aligned with UN Regulation No. 79 standards.
- **Hands-on Testing & Validation:** Actively engaged in feature evaluation by interfacing with the vehicle's CAN network through central mirror cameras using Vector tools (CANoe and CANape), executing rigorous proving ground tests, and interpreting complex data logs. Led validation and verification of ADAS features on Hardware-in-the-Loop (HiL) and Vehicle-in-the-Loop (ViL) platforms.
- **Risk Mitigation & Defect Resolution:** Led cross-functional defect resolution when boundary-condition testing revealed delayed trigger alerts; collaborated with subsystem engineers and suppliers to adjust sensor Field of View (FOV) parameters, carefully balancing detection improvements against false-positive risks (a SOTIF performance-limitation case). Managed defect tracking through Rational Team Concert (RTC/RQM).

Mechanical Engineer Internship

05/2014 – 08/2014

Temsa Global Adana, Türkiye

- Supported continuous improvement initiatives, defect tracking, and quality optimization directly on the manufacturing production line.
- Validated power and torque hardware requirements utilizing MATLAB/Simulink for electric bus powertrains.

EDUCATION

MSc in Sustainable Automotive Engineering

10/2017 – 09/2018

Warwick Manufacturing Group, Coventry, UK

- **Grade:** 2:1 (Merit)
- **Dissertation:** Experimental determination of variables in a thermal model of a wound stator, utilizing MATLAB, Simulink, and MotorCAD for complex system modelling.

BSc in Mechanical Engineering

08/2011 – 02/2017

Ege University, Izmir, Türkiye

- **Dissertation:** Manufactured and modelled a physical prototype 6-Degree of Freedom Stewart Simulation System. Designed hardware via SolidWorks, modelled movements in MATLAB, and built control algorithms in Arduino Uno.

TECHNICAL SKILLS

ADAS & Standards	Feature Concept to Production, Requirements Management, OEM Stakeholder Management, V-Model Validation, ASPICE, ISO 26262 (FuSa, trained), ISO 21448 (SOTIF, trained), UN Reg 79, Homologation, EuroNCAP Standards, Proving Ground Testing
Networks & Engineering Tools	Vector CANoe, CANape, standard automotive CAN, CAN FD, Ethernet, MATLAB, Simulink, CarMaker, Hardware-in-the-Loop (HiL)
Management & Requirements	RMDV2 DOORS, MKS Integrity, Rational Team Concert (RTC/RQM), Atlassian Stack (Jira, Confluence)
Software & Workflows	Python, OpenSCENARIO (OSC2) DSL, Efficient AI Workflows for Test Automation
Languages	English (Fluent/C2), Turkish (Native)